A Compressed Presentation of OpenAI's Proof of the Unique Games Theorem

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Abstract

We reconstruct the main argument of OpenAI's preprint *The Unique Games Theorem* [1]. Three earlier theorems are stated as inputs. The remaining argument constructs stable nonlinear noise, extracts constant-output certificates, decodes them using only local information, and contradicts parallel repetition on clean coordinates. We retain the conditioning and parameter order on which these steps depend. The aim is a short, continuous proof for a reader comfortable with finite-field linear algebra and Fourier analysis on finite groups who has not read the preprint. We modify the root construction of the noise gadget, simplify certificate extraction, and rewrite the genericity proof; these changes are identified and proved explicitly, and are listed, with smaller ones, under Attribution.

Status. This is an explanatory reconstruction. Its mathematical content received source-based reviews by AI agents (separate Claude instances and Codex); Codex had also helped write an earlier version. These reviews do not certify the preprint or constitute formal verification of this note, and no human expert has reviewed it. OpenAI's repository also contains a Lean 4 formalization of the source theorem, whose Comparator check passed when we ran it; that check concerns OpenAI's formal development, not this note or its changes (Appendix A). This note is not derived from the formalization. Fortnow's earlier exposition [6] presents the same proof at greater length; this note was prepared independently of it. A shorter proof, which simplifies several of the steps presented here, is given in [7]. The theorem below is the preprint's claim, whose proof argument we present. The root step, Lemma 4, has no counterpart in the source; it and our other departures from the source are listed under Attribution, at the end of Section 6.

1 The claim and three prior inputs

A unique game labels vertices from a finite alphabet. Each edge carries a permutation: a label at one endpoint specifies exactly one acceptable label at the other. Its value is the largest fraction of constraints satisfied by one labeling.

Theorem 1 (Main claim of [1])**.** *For every fixed $0 < \varepsilon, \delta < 1/2$, there is a deterministic polynomial-time reduction from satisfiability to explicit unweighted unique games on simple bipartite graphs that maps satisfiable formulas to YES games and unsatisfiable formulas to NO games, where*

$$\text{YES} : \operatorname{val}(G) \geq 1 - \varepsilon, \qquad \text{NO} : \operatorname{val}(G) \leq \delta.$$

The alphabet is a fixed binary vector space, and every constraint is a translation. The alphabet and the degree of the running-time polynomial may depend on the two target errors, but not on the input length.

All spaces and linear maps below are over $\mathbb{F}_2$, except where a larger finite field is specified. Uniform samples include zero. Separately specified random choices are independent unless stated otherwise. Intermediate games may have rational weights and repeated constraint occurrences.

Input 1: the parity gap. For every fixed positive rational ξ, Håstad's theorem [2, Theorem 5.4 and its proof] gives a deterministic polynomial-time reduction from satisfiability to weighted three-bit equations

$$x_{v_1} + x_{v_2} + x_{v_3} = b_e, \qquad \text{YES} : \text{OPT} \geq 1 - \xi, \qquad \text{NO} : \text{OPT} \leq \tfrac{1}{2} + \xi.$$

The rational weights have polynomial encoding length.

Input 2: Grassmann expansion. The binary Grassmann graph has the ℓ-dimensional subspaces of an ambient space as vertices; adjacency means an intersection of dimension $\ell - 1$. The retention of a nonempty set is the probability that a uniform neighbor of a uniform member stays in the set. For every fixed retention threshold $\zeta > 0$, there are $\alpha > 0$ and an integer $r_{\mathrm{s}} \geq 1$, depending only on ζ, such that every set with retention at least ζ has relative density at least α in an interval

$$\{L : A_0 \subseteq L \subseteq B_0\}, \qquad \dim A_0 + \operatorname{codim} B_0 \leq r_{\mathrm{s}}.$$

This holds for every sufficiently large ℓ, and then every sufficiently large ambient dimension. This is the expansion theorem of Khot–Minzer–Safra [3, Theorem 1.12].

Input 3: projection-game repetition. In a projection game, an answer of the first player determines one required answer of the second. Dinur–Steurer [4, Corollary 1.2] give

$$\operatorname{val}(G) \leq 1 - g \quad \Longrightarrow \quad \operatorname{val}(G^{\otimes n}) \leq (1 - g^2/16)^n. \tag{1}$$

This applies to finite weighted games and independent constraint samples. Each player may use its entire question tuple. The bound does not depend on the alphabet. Our games have a fixed projection for each question pair; repeated samples of the same constraint simply contribute their weights.

Idea and plan. The honest labeling applies one nonlinear map C, and the test perturbs its input by a special noise. The noise rarely changes the output of C, yet every linear family that sees enough of the output notices it with constant probability. Section 3 builds such noise by a race inside a recursive construction. At each level an output change survives with probability $1 - \theta$, so after s levels it survives with probability about $e^{-\theta s}$. A generic family of linear observations of rank r loses at most one dimension per level, and does so with probability at most $3r\theta$. Like a coupon collector, it therefore needs at least about $H_r/(3\theta)$ levels to die, where $H_r = 1 + \frac{1}{2} + \cdots + \frac{1}{r}$. The harmonic numbers diverge, so a large initial rank wins; the proof takes $s \approx H_{r_0}/(8\theta)$. The algebra needed is that, for a generic family, at most $3r$ of the θ^{-1} field lines are bad, each costing at most one dimension. A bad-line bound such as 2^r would make the corresponding sum converge, and this argument would fail. Stability gives completeness (Section 2). Conversely, detection and Input 2 turn a labeling that passes the test into constant-output certificates, and these give two local decoders with constant agreement (Section 4). In a NO instance, clean coordinates embed the repeated base game, and (1) makes such agreement impossible (Section 5). Section 6 removes weights and amplifies the gap.

Proof map. Four statements carry the argument; the notation is defined in Sections 2–5.

(i) *Gadget:* the contract (3), with r_* depending only on p_* (Section 3, Lemma 4).

(ii) *Completeness:* (6).

(iii) *Extraction:* for all cubes k beyond a threshold, any labeling that passes the test with probability at least 99/100 yields two local strategies with legal answers that agree with probability at least a constant $\gamma > 0$ (Proposition 5).

(iv) *Clean coordinates:* if $\mathrm{OPT}(\mathcal{E}) \le 4/5$, as in a NO instance with $\xi < 1/100$, every pair of local strategies with legal answers in the same experiment agrees with probability at most $\exp[-2^{-\dim W - r_s} k^{1/3}/3600]$ (Proposition 11).

The threshold, γ, $\dim W$ and r_s are fixed before k and do not depend on the instance or the labeling; this matters because the parity error, and hence the instance, is chosen after k. So (iii) and (iv) give the NO side of the gap (25), and (ii) gives its YES side. Section 6 turns the gap into Theorem 1.

2 From parity equations to shared tables

Distinct variables in each equation. Let $\mathcal{E}_0$ be the system given by Input 1. Replace every variable by 100 clones. In each equation, enumerate uniform clone choices conditioned on repeated occurrences of one variable using distinct clones. Split the original weight among these choices. Copying a global assignment to all clones preserves completeness.

For an arbitrary clone assignment put $t_v = \mathbb{E}_c(-1)^{x_{v,c}}$. Before conditioning, the equation's satisfaction probability is

$$\frac{1 + (-1)^{b_e} t_{v_1} t_{v_2} t_{v_3}}{2}.$$

Round each variable by the sign of its bias, breaking zero ties arbitrarily. An equation violated after rounding has satisfaction probability at most 1/2 before conditioning; any other equation has probability at most one. The probability of a forbidden clone collision is at most 3/100. Conditioning away an event of probability p changes the expectation of a $[0,1]$-valued variable by at most p. Write $\mathcal{E}$ for the processed system and ω_e for its weights. Then $\mathrm{OPT}(\mathcal{E}) \ge \mathrm{OPT}(\mathcal{E}_0)$ and

$$\mathrm{OPT}(\mathcal{E}) \le \frac{1 + \mathrm{OPT}(\mathcal{E}_0)}{2} + \frac{3}{100},$$

so in the NO case, for $\xi < 1/100$, $\mathrm{OPT}(\mathcal{E}) \le 4/5$.

In its equation–variable game, the first player receives an equation and returns a satisfying triple. The second receives a uniformly chosen one of its three variables and returns a bit. The second player's answers form a global assignment. On an equation this assignment violates, every legal triple disagrees with it in at least one position. Hence in the NO case

$$\mathrm{val}(G_{\mathcal{E}}) \le 1 - \frac{1 - \mathrm{OPT}(\mathcal{E})}{3} \le 14/15. \tag{2}$$

Distinct variables ensure that an equation–variable question pair specifies a unique position.

The noise gadget needed below. Section 3 constructs, for each $p_* > 0$, a threshold r_* such that for every sufficiently large ℓ divisible by 3 there are finite data

$$\begin{gathered} K = \mathbb{F}_2^{\ell} \le \mathcal{V}, \quad C : \mathcal{V} \to K, \quad a \sim \mu, \\ C(x+h) = C(x) + h \quad (h \in K), \qquad \Pr_{x \text{ uniform}, a}[C(x+a) \ne C(x)] \le p_*, \\ G \le \mathcal{V}^*,\ \dim(G|_K) \ge r_* \quad \Longrightarrow \quad \Pr_a[\exists g \in G : g(a) \ne 0] \ge 1/8. \end{gathered} \tag{3}$$

The noise law is rational. Crucially, r_* is chosen before ℓ. Fix a linear splitting $\mathcal{V} = K \oplus W$.

Questions and homogeneous answers. Sample $U = (e_1, \dots, e_k)$ independently with replacement. Put $L_U = \mathbb{F}_2^{1+2k}$, with coordinates $t_0, x_1, y_1, \dots, x_k, y_k$. The triple for equation e_i is

$$(x_i, y_i, b_{e_i} t_0 + x_i + y_i).$$

Vectors with $t_0 = 1$ are exactly the legal answers. Different positions retain separate answer coordinates, even when their variable identifiers repeat. A projected question O retains either an equation or one of its actual bits in each position; a singleton position records only the variable's identifier, not the equation or the slot it came from. This gives a surjective linear map $\pi : L_U \to L_O$ preserving t_0.

Table keys force sharing. A table is a linear map $P : L_Q \to \mathcal{V}$, or an affine map on the legal answers. In each position a table depends on the whole equation, on one of its variables, or on nothing. Its key is the coarsest support pattern (positions, types and identifiers) through which it factors, together with the induced affine map. Concretely, reduce it to its support. Delete zero-slope blocks; retain an equation when its two slopes are independent. A nonzero rank-one block has slopes $(v, 0), (0, v)$ or (v, v) and retains the first, second or third variable, respectively. In the third case adjust the intercept, using the third bit $z_i = b_{e_i} t_0 + x_i + y_i$:

$$v(x_i + y_i) = v z_i + b_{e_i} v t_0.$$

For a singleton block, retain it exactly when its slope is nonzero. The key records positions, question types and identifiers, and the full reduced affine map, including its intercept. Pulling a projected table back to U and reducing again gives the same key: the third-bit intercept correction occurs twice and cancels. Such keys therefore already occur among tables on equation tuples.

Take vertices to be key orbits under addition of a constant from K. Choose the representative whose intercept has K-component zero. Restore a table's output by adding its removed constant to the vertex label. Every labeling then defines functions satisfying

$$F_Q(P + h e_0^\top) = F_Q(P) + h, \qquad F_U(P'\pi) = F_O(P'). \tag{4}$$

Here $e_0^\top$ is the first-coordinate functional.

The test and completeness. Sample $U \sim \omega^{\otimes k}$, a uniform table P, noise $a \sim \mu$, and a uniform $l \in \mathbb{F}_2^m$, where $m = 1 + 2k$. Require

$$F_U(P) = F_U(P + a l^\top). \tag{5}$$

Equality of restored outputs is a translation constraint on the two vertex labels. The sampling probabilities are its rational weights. Call this weighted translation game $\mathcal{H}$. Loops and repeated edges are allowed for now.

Given a global assignment satisfying at least $1 - \xi$ of the equations, answer each retained variable by its global bit, and each retained equation by the global first two bits and the required third bit. Label a key representative by evaluating its table on this answer and applying C. Equivariance makes restoration consistent. With probability at least $1 - k\xi$, every equation in U is satisfied. All support answers are then projections of one legal vector z_U. The compared values are

$$C(P z_U), \qquad C(P z_U + a(l^\top z_U)).$$

Since $z_U \neq 0$, the first argument is uniform and $l^\top z_U$ is an independent fair bit. Consequently

$$\mathrm{val}(\mathcal{H}) \geq 1 - k\xi - p_*/2. \tag{6}$$

Why a gadget is needed. Without one, take $\mathcal{V} = K$, $C = \mathrm{id}$ and a uniform on K. For a legal answer z, nonzero since $t_0 = 1$, the evaluation labeling $F(P) = Pz$ is folded and passes (5) exactly when $a(l^\top z) = 0$. As $l^\top z$ is a fair bit independent of a, it accepts with probability exactly $\frac{1}{2} + \frac{1}{2|K|}$. The gadget's nonlinear C absorbs the noise, so acceptance approaches one by (6), while every linear family of large restriction rank still detects it. Soundness exploits this detection through Lemma 6.

3 The latent noise gadget

We prove the contract (3): for every $p_* > 0$ we construct a threshold r_*, and then, for every sufficiently large ℓ divisible by 3, finite data $K \le \mathcal{V}$, C and a rational law μ. Here x is uniform on $\mathcal{V}$, a has law μ, and the two are independent. All vector spaces and linear maps in this section are over $\mathbb{F}_2$ unless another field is specified.

3.1 One quadratic block

Take $F = \mathbb{F}_q$, where $q = 2^d$, and put $X = B_{\mathrm{blk}} = F^3$ and $d_{\mathrm{blk}} = 3d$. Identify their binary duals with F^3 using the nondegenerate pairing $\mathrm{Tr}(u \cdot v)$, where $\mathrm{Tr} : F \to \mathbb{F}_2$ is the field trace. Define

$$Q(x) = (x_2x_3, x_1x_3, x_1x_2), \qquad D(x,a) = Q(x+a) + Q(x) + Q(a), \qquad C_{\mathrm{blk}}(x,y) = y + Q(x).$$

For each one-dimensional F-subspace $A = Fv$ of F^3, let

$$D_A = \{w : v \cdot w = 0\}, \qquad U_A = \{(a, Q(a) + w) : a \in A,\ w \in D_A\}.$$

Expanding one coordinate, $(x_2+a_2)(x_3+a_3)+x_2x_3+a_2a_3 = x_2a_3+x_3a_2$, shows, by symmetry, that $D(x,a) = x \times a$ is the cross product; signs do not matter in characteristic two. Hence

$$v \cdot D(x, tv) = t\, v \cdot (x \times v) = 0, \qquad D(tv, t'v) = tt'\, v \times v = 0.$$

The second identity says that $Q|_A$ is additive, so U_A is a binary subspace of dimension d_{blk}. The trace annihilator of D_A is A: containment holds, and both have binary dimension d. The spaces U_A together span $X \oplus B_{\mathrm{blk}}$. Indeed, the coordinate lines supply all horizontal vectors, since Q vanishes there, and their vertical planes span B_{blk}.

The cancellation identity is

$$C_{\mathrm{blk}}((x,y) + (a, Q(a) + w)) + C_{\mathrm{blk}}(x,y) = w + D(x,a) \in D_A.$$

For a uniform update from U_A, this difference is uniform on D_A, even with the input and a fixed. There are q^2+q+1 field lines. Set $\theta = (q^2+q+1)^{-1}$. Removing the zero update, which has probability q^{-3} and never changes the output, gives

$$\Pr[\text{output changes under a uniform nonzero update in } U_A] = \frac{1 - q^{-2}}{1 - q^{-3}} = 1 - \theta. \tag{7}$$

Now fix a binary subspace $S \le B_{\mathrm{blk}}^*$ and a linear extension $z \mapsto (\gamma_z, z) \in X^* \oplus B_{\mathrm{blk}}^*$, before choosing A. Write $\gamma_z(x) = \mathrm{Tr}(g(z) \cdot x)$, where $g : S \to F^3$ is binary-linear. For $z \neq 0$,

$$(\gamma_z, z)|_{U_A} = 0 \quad \Longleftrightarrow \quad A = Fz \ \text{ and } \ z \cdot g(z) = \sqrt{z_1z_2z_3}. \tag{8}$$

The vertical updates first force $z \in A$. The remaining condition is

$$\mathrm{Tr}(t\, g(z) \cdot z + t^2 z_1z_2z_3) = 0 \quad \text{for every } t \in F.$$

Since $\mathrm{Tr}(t^2c) = \mathrm{Tr}(t\sqrt{c})$, nondegeneracy of the trace proves (8). Square roots here are unique because squaring is an automorphism of a finite field of characteristic two.

3.2 Generic spaces work for every extension

Call an r-dimensional binary subspace $S \leq F^3$ *generic* if no two distinct nonzero elements are F-proportional and, for every binary-linear $g : S \to F^3$, the alignment equation in (8) has at most $3r$ nonzero solutions. For each fixed r, a uniform r-dimensional subspace is generic with probability tending to one as q tends to infinity. We prove this uniform statement, including its quantification over all g.

Use independent uniform parameters $y_{ij} \in F$, and write

$$z_i(t) = \sum_{j=1}^{r} y_{ij}^2 t_j, \qquad c_t = \prod_{i=1}^{3} \left(\sum_{j=1}^{r} y_{ij} t_j \right), \qquad t \in \mathbb{F}_2^r.$$

Thus $c_t = \sqrt{z_1(t) z_2(t) z_3(t)}$. Squaring preserves uniformity. The map $t \mapsto z(t)$ fails to be injective with probability at most $(2^r - 1)/q^3$; conditional on injectivity its image is a uniform r-space. Two distinct nonzero binary masks are linearly independent over F, so their images are independent uniform vectors. Their probability of being nonzero and proportional is $O(q^{-2})$. A finite union over pairs proves the first genericity condition.

Fix a set $E \subseteq \mathbb{F}_2^r \setminus \{0\}$ with $|E| > 3r$, and form

$$M_{t,(i,j)} = t_j z_i(t) \qquad (t \in E,\ 1 \leq i \leq 3,\ 1 \leq j \leq r).$$

An extension aligning at all masks in E exists exactly when $c \in \mathrm{col}(M)$: the unknowns are the three coordinates of the r values defining g. This single system handles every observer.

First regard the parameters as indeterminates, and put

$$\mathbb{L}_0 = \mathbb{F}_2(y_{ij}^2) \subset \mathbb{L}_1 = \mathbb{F}_2(y_{ij}).$$

Lemma 2 (A cubic outside a quadratic span)**.** *If $E \subseteq \mathbb{F}_2^r \setminus \{0\}$ and $|E| > 3r$, then c is not in the column space of M, even over an algebraic closure of $\mathbb{L}_1$.*

Proof. The squarefree monomials in all the variables y_{ij} form a basis of $\mathbb{L}_1$ over $\mathbb{L}_0$. In particular, the monomials $y_{1j}y_{2k}y_{3l}$ are linearly independent over $\mathbb{L}_0$. To see this, clear denominators. Coefficients then have only even exponents, whereas these monomials have distinct exponent-parity patterns, so cancellation is impossible. The choice of the smaller coefficient field is essential.

Suppose that c belongs to the column space of M over such a closure. Every row annihilator of M with entries in $\mathbb{L}_0$ then annihilates c. Independence of the preceding monomials forces it to annihilate each vector $(t_j t_k t_l)_{t \in E}$ separately. By the annihilator characterization of a column space, all these vectors belong to $\mathrm{col}_{\mathbb{L}_0}(M)$. Let V_j be the $\mathbb{L}_0$-span of evaluations on E of positive-degree monomials of degree at most j. Repeated indices cover degrees one and two, giving

$$V_3 \subseteq \mathrm{col}_{\mathbb{L}_0}(M) \subseteq V_2 \subseteq V_3.$$

Multiplication by any input coordinate takes V_2 into V_3, so it preserves this common space. The space therefore contains every positive-degree monomial. It contains the indicator of each point of E: the usual Boolean point-indicator polynomial has a positive coordinate factor because the point is nonzero. Consequently the column space has dimension $|E| > 3r$, a contradiction. □

Now specialize. Let ϱ_M be the formal rank of M. There is a nonzero augmented $(\varrho_M + 1)$-minor of $[M \mid c]$, whereas every unaugmented minor of that size vanishes identically. Its polynomial degree is at most $2\varrho_M + 3 \leq 6r + 3$. Thus any specialization admitting an alignment on E zeros this polynomial, including specializations where the rank drops. A nonzero polynomial vanishes at independent uniform field elements with probability at most its total degree divided by q: induct on the number of variables, using the leading coefficient and the one-variable root

bound. Union over sets E of size $3r+1$ now finishes the proof. Their number depends only on r; if no such sets exist the condition is automatic. Conditioning on injectivity costs a factor tending to one.

Here is the rank estimate we will use. Put $H_0 = 0$ and $H_r = \sum_{i=1}^{r} 1/i$, and let

$$r_A = \dim \operatorname{im}\big[z \mapsto (\gamma_z, z)|_{U_A}\big], \qquad r = \dim S.$$

For generic S, (8) gives

$$r - r_A \le 1, \qquad \Pr_A[r_A < r] \le 3r\theta, \qquad \mathbb{E}_A(H_r - H_{r_A}) \le 3\theta. \tag{9}$$

The first bound uses the absence of proportional pairs. For arbitrary S we still have

$$\Pr_A[r_A < r] \le 2^r\theta, \qquad H_r - H_{r_A} \le H_r. \tag{10}$$

3.3 Recursion reduces the nonlinear error

Construct spaces $R_t \le P_t$, surjections $\lambda_t : R_t \to B_{\mathrm{blk}}$, and maps $C_t : P_t \to B_{\mathrm{blk}}$ satisfying

$$C_t(u+h) = C_t(u) + \lambda_t(h) \qquad (h \in R_t). \tag{11}$$

At height zero all spaces are B_{blk} and both maps are the identity. At each subsequent height use one child for each pair in

$$\mathcal{I} = \{(A, J) : J : B_{\mathrm{blk}} \xrightarrow{\sim} U_A\}, \qquad P_t = \prod_{\mathcal{I}} P_{t-1}, \qquad C_t(u) = C_{\mathrm{blk}}\left(\sum_{(A,J)} JC_{t-1}(u_{A,J})\right).$$

All isomorphisms here are binary. On the product of child translation spaces define the surjections

$$L(h) = (\lambda_{t-1}(h_i))_i, \qquad \mathcal{A}((b_i)_i) = \sum_i J_i b_i = (\mathcal{A}_X((b_i)_i), \mathcal{A}_B((b_i)_i)).$$

The second map is onto because the update spaces span $X \oplus B_{\mathrm{blk}}$. Set

$$R_t = \ker(\mathcal{A}_X L), \qquad \lambda_t = (\mathcal{A}_B L)|_{R_t}.$$

This last map is onto. Such translations change only the vertical input of the block, proving (11). In particular, a uniform input has uniform output.

The noise chooses a uniform child at every level and adds an independent uniform element of B_{blk} at the selected leaf. Each line has the same number of isomorphisms, so a uniform child can be exposed as a uniform line followed by a fresh uniform isomorphism.

We need uniformity even after conditioning on a child's output change. For any fixed child perturbation a, write $\Delta_a(u) = C_{t-1}(u+a) + C_{t-1}(u)$. Then

$$(\Delta_a, C_{t-1})(u+h) = (\Delta_a(u), C_{t-1}(u) + \lambda_{t-1}(h)).$$

Translation preserves the uniform input law. Surjectivity of λ_{t-1} therefore makes the child output uniform conditional on Δ_a and a. The other child outputs remain independent and uniform. Since $\mathcal{A}$ is onto, the block input is uniform conditional on the chosen child and its change.

All children are identical copies, so the law of that change is independent of the chosen child index. Given a nonzero change and the chosen line, the fresh uniform J sends the change to a uniform nonzero element of U_A. Equation (7) thus gives the exact recursion

$$p_0 = 1 - q^{-3}, \qquad p_t = (1 - q^{-3})(1-\theta)^t, \tag{12}$$

where p_t is the probability that C_t changes under its noise.

3.4 Linear rank survives the adaptive path

A lift of $S \le B_{\mathrm{blk}}^*$ is a linear map $z \mapsto \psi_z \in P_t^*$ with $\psi_z|_{R_t} = z\lambda_t$. We show that a lift determines a single extension at the current block, before a child is chosen. Restrict ψ_z to $\prod_{\mathcal{I}} R_{t-1}$. It vanishes on $\ker L$, hence descends to a functional χ_z on $\prod_{\mathcal{I}} B_{\mathrm{blk}}$. The functional $\chi_z - z\mathcal{A}_B$ vanishes on $\ker \mathcal{A}_X$. Since $\mathcal{A}_X$ is onto, there is a unique $\gamma_z \in X^*$ such that

$$\chi_z - z\mathcal{A}_B = \gamma_z \mathcal{A}_X, \qquad \phi_{A,J}(z) = J^*((\gamma_z, z)|_{U_A}). \tag{13}$$

Uniqueness makes γ_z linear in z, and common to all children.

For $h \in R_{t-1}$ in child (A, J) this gives $\psi_z(h) = \phi_{A,J}(z)(\lambda_{t-1}h)$. For a chosen child, let $S' = \operatorname{im} \phi_{A,J}$. Choose a linear section $\iota : S' \to S$ and restrict $\psi_{\iota(z')}$ to that child's input. This is a lift of S'. Repeat along the noise path, choosing sections by a fixed rule from the revealed data. If a continuing functional detects the leaf noise, its ancestor detects the original noise, because that noise is supported only in the selected child at every level.

Choose $r_0 \ge 1$ and $0 < \zeta_0 < 1$ with

$$H_{r_0} 2^{r_0} \zeta_0 \le 1.$$

Choose q large enough that $d_{\mathrm{blk}} \ge r_0$ and that, at every rank from 1 to r_0, the proportion of nongeneric subspaces is at most ζ_0. Start from any generic r_0-dimensional space $S_0 \le B_{\mathrm{blk}}^*$ together with a lift of it.

Let $\mathcal{F}_j$ contain the full path history just before choosing the next line and isomorphism. The current space, lift, and common extension are $\mathcal{F}_j$-measurable. After the line A_j is revealed, the next rank is already fixed:

$$\rho_{j+1} = \dim E_j, \qquad E_j = \operatorname{im}[z \mapsto (\gamma_z, z)|_{U_{A_j}}].$$

Only then expose J_j. The space $S_{j+1} = J_j^* E_j$ is uniform among subspaces of dimension ρ_{j+1}. Its lift may depend on J_j; this is harmless because genericity holds for every lift on a generic space.

The rest is a probability statement, which we isolate.

Lemma 3 (The race)**.** *Let $\theta > 0$, and let $r_0 = \rho_0 \ge \rho_1 \ge \cdots \ge 0$ be integers with $r_0 \ge 1$ and $B_0, B_1, \ldots \in \{0, 1\}$, all adapted to a filtration $(\mathcal{F}_j)$. Suppose that*

$$\mathbb{E}B_j \le \zeta_0, \qquad \mathbb{E}[H_{\rho_j} - H_{\rho_{j+1}} \mid \mathcal{F}_j] \le 3\theta + H_{r_0} 2^{r_0} \theta B_j, \qquad H_{r_0} 2^{r_0} \zeta_0 \le 1.$$

Then $s = \lfloor H_{r_0}/(8\theta) \rfloor$ satisfies $\Pr[\rho_s > 0] \ge \frac{1}{2}$.

Proof. Taking expectations, each step lowers $\mathbb{E}H_{\rho_j}$ by at most 4θ. So $\mathbb{E}H_{\rho_s} \ge H_{r_0} - 4\theta s \ge H_{r_0}/2$, while $H_{\rho_s} \le H_{r_0} \mathbf{1}_{\{\rho_s > 0\}}$. □

The bound on $\mathbb{E}B_j$ is unconditional. The exceptional term is removed only after averaging over the history, not under conditioning on it.

To apply the lemma, let B_j indicate a positive-rank nongeneric current space. Then $B_0 = 0$ and $\mathbb{E}[B_{j+1} \mid \mathcal{F}_j, A_j] \le \zeta_0$, because S_{j+1} is uniform of its dimension. The conditional loss bound follows from (9)–(10). So $\rho_s > 0$ with probability at least $1/2$. The uniform leaf symbol has not yet been exposed. A family of rank $\rho_s > 0$ detects it with probability $1 - 2^{-\rho_s} \ge 1/2$. Ancestor detection therefore shows that every lift of a generic r_0-dimensional space detects the noise with probability at least $1/4$.

3.5 A random root and one quotient

Fix a binary basis of F^3 and put $K = B_{\mathrm{blk}} = \mathbb{F}_2^\ell$. The following lemma completes the gadget; its random root automorphism plays the role of the fresh isomorphisms J at every level.

Lemma 4 (Root step)**.** *Let r_0, ζ_0, d and s be as in Section 3.4, so that (i) every lift on P_s of a generic r_0-dimensional subspace of B^*_{blk} detects the depth-s noise with probability at least $1/4$, and (ii) a uniform r_0-dimensional subspace of B^*_{blk} is generic with probability at least $1-\zeta_0 \geq 1/2$. Then, with output dimension $\ell = d_{\mathrm{blk}} = 3d$, there are a finite binary space $\mathcal{V}$ containing K, a map $C : \mathcal{V} \to K$ and a rational law μ on $\mathcal{V}$ such that: (a) $C(x+h) = C(x)+h$ for all $x \in \mathcal{V}$ and $h \in K$; (b) for uniform x and independent $a \sim \mu$, $\Pr[C(x+a) \neq C(x)] = p_s$, the nonlinear error of the depth-s construction; and (c) every $G \leq \mathcal{V}^*$ with $\dim(G|_K) \geq r_0$ satisfies $\Pr_{a\sim\mu}[\exists g \in G : g(a) \neq 0] \geq 1/8$.*

Proof. Here (i) is the conclusion of Section 3.4; in (ii), $\zeta_0 \leq 1/2$ because $H_{r_0} 2^{r_0} \zeta_0 \leq 1$ and $r_0 \geq 1$. Take one copy of the depth-s construction for every binary automorphism $\tau \in GL(B_{\mathrm{blk}})$:

$$P = \prod_\tau P_s, \qquad R = \prod_\tau R_s, \qquad \widetilde{C}((u_\tau)) = \sum_\tau \tau\, C_s(u_\tau), \qquad \Lambda((h_\tau)) = \sum_\tau \tau\, \lambda_s(h_\tau).$$

By (11), $\widetilde{C}(u+h) = \widetilde{C}(u) + \Lambda(h)$ for $h \in R$, and $\Lambda : R \to K$ is onto because λ_s and each τ are. Take the single quotient

$$\mathcal{V} = P/\ker\Lambda, \qquad R/\ker\Lambda \simeq K.$$

Then $\widetilde{C}$ descends to $C : \mathcal{V} \to K$ with $C(x+h) = C(x) + h$ for $h \in K$, and K is an actual subspace of $\mathcal{V}$. The noise draws a uniform τ and, independently of τ, a depth-s noise sample $\xi_s \in P_s$; it places ξ_s in copy τ and passes to the quotient. A uniform input on P pushes forward to a uniform input on $\mathcal{V}$, since all cosets of $\ker\Lambda$ have the same size. Since τ is bijective, the output change event is preserved exactly, and its probability is still p_s. This proves (a) and (b); μ is rational, being a pushforward of uniform choices on finite sets.

For (c), fix $G \leq \mathcal{V}^*$ with $\dim(G|_K) \geq r_0$. It is fixed before the noise, and so before τ. Put $S = G|_K$ and fix an r_0-dimensional subspace $T \leq S$. Let $\varpi : P \to \mathcal{V}$ be the quotient map; on R it is Λ under the identification $R/\ker\Lambda \simeq K$. Thus for $g \in G$ and $h \in R_s$ in copy τ, $g(\varpi\, \iota_\tau h) = (g|_K)\tau\lambda_s(h)$, where ι_τ includes factor τ into P. The map $g \mapsto (g|_K)\tau$ sends G onto $\tau^* S$, which contains $\tau^* T = \{z\tau : z \in T\}$. For uniform τ, the space $\tau^* T$ is uniform among the r_0-dimensional subspaces of B^*_{blk}, since $GL(B_{\mathrm{blk}})$ acts transitively on them. By (ii) it is generic with probability at least $1-\zeta_0$.

Fix such a τ with $\tau^* T$ generic, and a linear section $\varsigma : \tau^* T \to G$ of $g \mapsto (g|_K)\tau$. The functionals $\psi_z = \varsigma(z) \circ \varpi \circ \iota_\tau$ satisfy $\psi_z|_{R_s} = z\lambda_s$, so they form a lift of $\tau^* T$. Since $a = \varpi\, \iota_\tau \xi_s$ with ξ_s drawn independently of τ, by (i) this lift detects ξ_s with probability at least $1/4$. Each ψ_z is the pullback of $\varsigma(z) \in G$, so $\psi_z(\xi_s) = \varsigma(z)(a)$, and G then detects a. Averaging over τ gives detection probability at least $(1-\zeta_0)/4 \geq 1/8$. □

Parameter order and finite data. Choose r_0 large enough that $e^{1-H_{r_0}/8} \leq p_*$; this is possible because the harmonic sums diverge. Set $r_* = r_0$ and choose ζ_0. The field degree d may then be any integer large enough that $3d \geq r_0$ and that, at every rank from 1 to r_0, the proportion of nongeneric subspaces is at most ζ_0. Finally set $\ell = d_{\mathrm{blk}} = 3d$, θ and s:

$$p_* \;\longrightarrow\; r_0 = r_* \;\longrightarrow\; \zeta_0 \;\longrightarrow\; d \text{ (any large enough)} \;\longrightarrow\; \ell = 3d,\ \theta,\ s.$$

Equation (12) gives

$$p_s \leq e^{-\theta s} \leq e^{\theta - H_{r_0}/8} \leq p_*.$$

Lemma 4 now gives the contract (3) with $r_* = r_0$, for every sufficiently large ℓ divisible by 3.

4 From high acceptance to private decoders

Fix p_* and its gadget threshold r_*. Set $\eta = 2^{-r_*}/16$. Apply Input 2 at retention $\eta/4$ to obtain α, r_{s}. Choose ℓ divisible by 3 and large enough for that input and the gadget, and so that

$$2^{-(\ell - r_{\mathrm{s}})} < \alpha/8, \qquad 2^{-\ell} < \eta/4. \tag{14}$$

Now fix the gadget and splitting $\mathcal{V} = K \oplus W$. Only k grows from now on. Write a table as $P = (M, T)$ with values in $K \oplus W$.

Extraction concerns the following experiment.

The precise experiment. Sample $U \sim \omega^{\otimes k}$. Independently in each position, retain the equation with probability $1 - \beta$, or a uniform one of its variables with probability $\beta = k^{-2/3}$. This gives O and $\pi : L_U \to L_O$. Sample independent uniform maps

$$A : K \to \mathbb{F}_2^{r_{\mathrm{s}}}, \qquad T_O : L_O \to W, \qquad M_O : L_O \to K, \qquad S_O = AM_O.$$

The messages are exactly

$$(U, A, T_O\pi, S_O\pi), \qquad (O, A, T_O, S_O). \tag{15}$$

Neither player receives M_O or π. Each may use the fixed function family and independent private randomness, but only its own message.

Proposition 5 (Extraction)**.** *There are k_0 and $\gamma > 0$, depending only on the fixed auxiliary data, such that the following holds for every parity instance and every cube $k \geq k_0$. If a labeling of $\mathcal{H}$ passes (5) with probability at least $99/100$, then there are such strategies, returning legal answers, with*

$$\Pr[\pi a_U = a_O] \geq \gamma. \tag{16}$$

The strategies need not be efficient.

The rest of this section proves the proposition. Choose k_0 so that every $k \geq k_0$, with $m = 1+2k$, gives $2^{-m} \leq \eta/4$, makes $\ell + m$ exceed Input 2's ambient-dimension threshold at retention $\eta/4$, and makes $v_k \leq \gamma_1$ in (21) and (22) below; this is possible because $v_k \to 0$ as $k\beta^2 = k^{-1/3} \to 0$.

4.1 Replacing the noise and extracting constant outputs

For a function on a finite binary space, put $\widehat{f}(\phi) = \mathbb{E}_x f(x)(-1)^{\phi(x)}$. A nonzero character averages to zero: translation by a point where it is one reverses its sign. Thus characters form an orthonormal basis, Fourier inversion holds, and Parseval gives $\sum_\phi |\widehat{f}(\phi)|^2 = \mathbb{E} f^2$. Also

$$\mathbf{1}_{u=v} = \mathbb{E}_{\sigma \in K^*} (-1)^{\sigma(u+v)}.$$

For a noise law ν on $\mathcal{V}$ and a function F from tables $L_U \to \mathcal{V}$ to K, let $\mathrm{acc}_\nu(F) = \Pr[F(P) = F(P + al^\top)]$, with P and l uniform and $a \sim \nu$. Let ν_K be uniform on K.

Lemma 6 (Noise comparison)**.** $\mathrm{acc}_{\nu_K}(F) \geq 2^{-r_*}\big(8\,\mathrm{acc}_\mu(F) - 7\big)$.

Proof. Expand the equality indicator over $\sigma \in K^*$, and each $(-1)^{\sigma F}$ in characters $g = (g_1, \ldots, g_m) \in (\mathcal{V}^*)^m$ of the table space. Orthogonality over P and averaging over l give

$$\mathrm{acc}_\nu(F) = \mathbb{E}_\sigma \sum_g |\widehat{F_\sigma}(g)|^2 \lambda_\nu(g), \qquad \lambda_\nu(g) = \Pr_{a \sim \nu}[g_i(a) = 0 \text{ for all } i] \geq 0,$$

and the squared coefficients sum to one for each σ. Call g low when the restrictions $g_i|_K$ span dimension less than r_*, and let m_{low} be the average mass on low frequencies. For the others,

the family $\mathrm{span}\{g_i\}$ has restriction to K of dimension at least r_*, and some element of it is nonzero at a exactly when some g_i is. So detection gives $\lambda_\mu \le 7/8$. Hence

$$\mathrm{acc}_\mu \le m_{\mathrm{low}} + \tfrac{7}{8}(1 - m_{\mathrm{low}}), \qquad \text{that is,} \qquad m_{\mathrm{low}} \ge 8\,\mathrm{acc}_\mu - 7.$$

On the low frequencies $\lambda_{\nu_K} = 2^{-\dim \mathrm{span}(g_i|_K)} \ge 2^{-r_*}$. □

The bound is linear, so it averages over U. Test acceptance 0.99 therefore gives average uniform-K acceptance at least $0.92 \cdot 2^{-r_*} = 14.72\eta$. This noise leaves T fixed. Since acceptance at fixed (U, T) lies in $[0, 1]$, a fraction at least $(14.72\eta - 4\eta)/(1 - 4\eta) > 10\eta$ of these pairs satisfies

$$\Pr_{M,a\in K,l}[F_U(M, T) = F_U(M + al^\top, T)] \ge 4\eta. \tag{17}$$

A constant-output slice. The *step retention* of a set of matrices is the probability that the uniform rank-one step $M \mapsto M + al^\top$, zero step included, keeps a uniform member inside it. For any function on matrices, its acceptance is the fiber-size-weighted average of the step retentions of its fibers.

Lemma 7 (Constant-output slice)**.** *Let ℓ, and then $\ell + m$, exceed the thresholds of Input 2 at retention $\eta/4$, and let $2^{-\ell}, 2^{-m} \le \eta/4$. Let f be a function on $\mathrm{Hom}(\mathbb{F}_2^m, K)$ with values in an arbitrary set, and suppose that the fiber of a value u has step retention at least η. Then there are a row map $A_* : K \to \mathbb{F}_2^{r_s}$, a value S_*, and at most r_s column conditions $Mq_j = t_j$ that cut out a nonempty slice of the row fiber $A_*M = S_*$ on which $f = u$ with density at least α.*

Proof. The step is zero with probability $p_{\mathrm{zero}} = 2^{-\ell} + 2^{-m} - 2^{-\ell-m} \le \eta/2$ by hypothesis. Conditioning on nonzero factors leaves retention at least $\eta/2$.

Identify M with its graph

$$\Gamma_M = \{(x, M^\top x) : x \in \mathbb{F}_2^\ell\}, \qquad \dim(\Gamma_M \cap \Gamma_N) = \ell - \mathrm{rank}(M - N).$$

These form the matrix chart in the Grassmann graph on $\mathbb{F}_2^\ell \oplus \mathbb{F}_2^m$. A chart vertex has $(2^\ell - 1)(2^m - 1)$ chart neighbors and $(2^\ell - 1)(2^{m+1} - 2)$ neighbors in total. For the latter count, choose a hyperplane in the vertex and a different one-dimensional extension in the quotient. For the former, use the unique nonzero rank-one factorization over $\mathbb{F}_2$. Exactly half the neighbors are in the chart. The lifted fiber has Grassmann retention at least $\eta/4$.

Input 2 gives relative density α in an interval. Intersecting it with the chart cannot lower density, since the entire fiber is already in the chart. A basis (d_i, s_i) of A_0 and a basis (t_j, q_j) of $B_0^\perp$, for the dot product on $\mathbb{F}_2^\ell \oplus \mathbb{F}_2^m$ under which $\Gamma_M^\perp = \{(Mv, v)\}$, express the interval's intersection with the chart as the nonempty slice

$$\{M : d_i^\top M = s_i^\top,\ Mq_j = t_j\}, \qquad i \le n_{\mathrm{row}},\ j \le n_c,\ n_{\mathrm{row}} + n_c \le r_s. \tag{18}$$

On at least an α fraction of it, f equals u. Its row conditions say $A_*M = S_*$, where A_* has the rows $d_i^\top$ padded with zeros, and there are at most r_s column conditions. □

The lemma needs no folding assumption, and values outside the fiber play no role. The constant output, rather than a general affine target, is all that is used below.

Good row advice occurs often. For fixed (U, T) write $f(M) = F_U(M, T)$. Call a nonempty row fiber $AM = S$ good if at most r_s column conditions cut out a nonempty slice of it on which one value of f has density at least α.

Lemma 8 (Good row advice is frequent)**.** *Assume the dimension hypotheses of Lemma 7. If (U, T) satisfies (17), the union $\mathcal{R}$ of all good row fibers, over all choices of A and S, has measure at least η.*

Proof. Suppose that $\mathcal{R}$ has measure less than η. Give each matrix in $\mathcal{R}$ its own new value outside K, and call the result f'. A rank-one step can lose a match only if an endpoint lies in $\mathcal{R}$. Both endpoints are uniform, so f' has acceptance at least $4\eta - 2\eta = 2\eta$. Each new value has a one-point fiber, whose step retention is exactly $p_{\text{zero}} \le \eta/2$. If every fiber of a value in K had step retention below η, the averaging identity would give acceptance below $\eta + p_{\text{zero}} < 2\eta$. So some $u \in K$ has a fiber of step retention at least η under f'. Lemma 7 gives a row fiber $A_*M = S_*$ and a slice of it, cut out by at most r_{s} column conditions, on which $f' = u$ with density at least α. Wherever $f' = u$, also $f = u$. So that row fiber is good, and lies in $\mathcal{R}$. But on $\mathcal{R}$ the function f' takes no value in K, a contradiction. □

For each matrix in $\mathcal{R}$, at least one of the $2^{\ell r_{\text{s}}}$ possible row maps witnesses goodness. Averaging also over the good pairs gives, under a uniform full table and independent uniform A,

$$\Pr[(U, A, T, AM) \text{ is good row advice}] \ge g_0 := 10\eta^2 2^{-\ell r_{\text{s}}}. \tag{19}$$

Select a certificate deterministically from each good advice, using a fixed order and the function family. Put $Z = \operatorname{span}\{q_j\}$. If every vector of Z has first bit zero, translations $M \mapsto M + he_0^\top$ with $h \in \ker A$ preserve the slice and translate its output by h. Every value then has density at most $2^{-\dim\ker A} \le 2^{-(\ell - r_{\text{s}})} < \alpha/8$, a contradiction. Therefore

$$\mathcal{A} = Z \cap \{v : e_0^\top v = 1\}, \qquad 1 \le |\mathcal{A}| \le 2^{r_{\text{s}}}. \tag{20}$$

The first player samples uniformly from this list, and gives any legal answer on advice that is not good. This uses only its own message.

4.2 Transferring a positive expectation

Fix U. Let $\mathcal{L}_{\text{pad}}$ be the law of the padded full table $(M_O\pi, T_O\pi)$, with its intercept, after averaging over the projection choices, and let $\mathcal{L}_{\text{unif}}$ be the law of a uniform table on L_U.

Lemma 9 (Padded tables look uniform)**.** *With $n = |\mathcal{V}|$, the total variation distance satisfies*

$$v_k = \|\mathcal{L}_{\text{pad}} - \mathcal{L}_{\text{unif}}\|_{\text{TV}} \le \frac{1}{2}\sqrt{\Big(1 + \frac{n-1}{3}\beta^2\Big)^k - 1}. \tag{21}$$

The same bound holds jointly with an independent uniform A.

Proof. Blocks are independent. A retained equation has uniform slopes. In a singleton block, the slope law Q_0 is the equally weighted mixture of $(a,0), (0,a), (a,a)$ for uniform $a \in \mathcal{V}$. Let P_0 be uniform on $\mathcal{V}^2$. The ratio Q_0/P_0 is n at zero, $n/3$ at the $3(n-1)$ other points of these three sets, and zero elsewhere, so $\chi^2(Q_0\|P_0) := \mathbb{E}_{P_0}(Q_0/P_0 - 1)^2 = (n-1)/3$. Mixing with weight β multiplies the centered ratio by β, and $1 + \chi^2$ multiplies over independent blocks. Cauchy–Schwarz bounds total variation by half the square root of χ^2. The intercept is covered too: an independent uniform intercept stays uniform after adding third-bit corrections that depend on the slopes. □

The lemma does *not* compare joint laws that include π. Since $k\beta^2 = k^{-1/3}$, the bound tends to zero.

Under the unrestricted uniform law, let J mean that row advice is good and the chosen column conditions hold; let E mean that the output equals the chosen constant. The conditions impose at most ℓr_{s} extra scalar equations on a nonempty row fiber. Hence

$$\Pr(J) \ge g_0 2^{-\ell r_{\text{s}}}, \qquad \mathbb{E}[\mathbf{1}_J(\mathbf{1}_E - \alpha/2)] \ge (\alpha g_0/2)2^{-\ell r_{\text{s}}}.$$

This score depends only on the full table, U, A, and the deterministic certificate selected from its row advice. Its range has length one, so (21) changes its expectation by at most $v_k \le \gamma_1$. Hence, in the projected experiment its expectation is at least

$$\gamma_1 := (\alpha g_0/4)2^{-\ell r_{\mathrm{s}}} > 0. \tag{22}$$

Only now condition on $D = (U, O, \pi, A, T_O, S_O)$. The first player's certificate is fixed, while M_O is uniform on its row fiber. By sharing, the selected output equality is an equality for F_O. Let p_D be the probability of its projected column slice and ρ_D the constant-output density there; take them to be zero when no such slice exists. The conditional score is $p_D(\rho_D - \alpha/2)$. It is nonpositive unless the slice is nonempty and $\rho_D \ge \alpha/2$, and is always at most one. Consequently these useful data occur with probability at least γ_1. No witness has been sent to the second player.

4.3 A decoder that does not know the certificate

The second player answers with a general decoding rule. Let $H \le K$, $d_H = \dim H$ and $X = \mathrm{Hom}(L_O, H)$, and let $g : X \to K$ be folded: $g(N + he_0^\top) = g(N) + h$ for $h \in H$. Identify

$$X^* = H^* \otimes L_O, \quad \langle \varphi \otimes v, N\rangle = \varphi(Nv), \quad (\varphi \otimes v)(h') = \varphi(h')v.$$

The decoder chooses $\sigma \in K^*$ uniformly, puts $f_\sigma = (-1)^{\sigma g}$, and samples $\Phi \in X^*$ with probability $|\widehat{f}_\sigma(\Phi)|^2$; Parseval normalizes this law. If $\sigma|_H \ne 0$, it returns $\Phi(h_\sigma)$ for a fixed $h_\sigma \in H$ with $\sigma(h_\sigma) = 1$. Otherwise it returns a fixed legal answer. In plain terms, a dense constant-output slice forces Fourier mass onto frequencies supported on the column directions that define the slice, and folding turns such a frequency into a legal answer.

Lemma 10 (Private decoding)**.** *The decoder always returns a legal answer. Let $Z' \le L_O$ have dimension s. If on some nonempty slice $N|_{Z'} = \kappa$ one value of g has density $\rho \ge 2^{-d_H}$, then with probability at least $2^{-d_H s}(\rho - 2^{-d_H})^2$ the decoder returns an answer in Z'.*

Proof. Folding gives $f_\sigma(N + h_\sigma e_0^\top) = -f_\sigma(N)$, so every sampled frequency satisfies $\langle \Phi, h_\sigma e_0^\top\rangle = 1$, that is, $\Phi(h_\sigma)$ has first coordinate one. Let $C_\sigma = \mathbb{E}[f_\sigma \mid N|_{Z'} = \kappa]$. The character identity for the dense value gives $\mathbb{E}_\sigma|C_\sigma| \ge \rho$. The characters vanishing on H form a 2^{-d_H} fraction, so $\mathbb{E}_\sigma[\mathbf{1}_{\sigma|_H \ne 0}|C_\sigma|] \ge \rho - 2^{-d_H}$. The maps vanishing on Z' have annihilator $\mathcal{F}_{\mathrm{col}} = H^* \otimes Z'$, of size $2^{d_H s}$. Averaging a character over the affine slice retains precisely these frequencies, so for any point N_* of the slice

$$C_\sigma = \sum_{\Phi \in \mathcal{F}_{\mathrm{col}}} \widehat{f}_\sigma(\Phi)(-1)^{\langle \Phi, N_*\rangle}.$$

Two applications of Cauchy–Schwarz give

$$\Pr[\sigma|_H \ne 0,\ \Phi \in \mathcal{F}_{\mathrm{col}}] = \mathbb{E}_\sigma\Big[\mathbf{1}_{\sigma|_H \ne 0} \sum_{\Phi \in \mathcal{F}_{\mathrm{col}}} |\widehat{f}_\sigma(\Phi)|^2\Big] \ge 2^{-d_H s}\big(\rho - 2^{-d_H}\big)^2.$$

On this event Φ maps H into Z', so the answer lies in Z'. □

The second player fixes a representative M_0 of its row fiber, with $AM_0 = S_O$, and applies the decoder to $g(N) = F_O(M_0 + N, T_O)$ with $H = \ker A$, using only its own message; g is folded by (4). For useful data D put $Z' = \pi Z$, so $s \le r_{\mathrm{s}}$. Since the slice is nonempty, $\kappa(\pi q_j) := t_j - M_0 \pi q_j$ defines a linear map $Z' \to H$. The transferred slice $N|_{Z'} = \kappa$ has a constant output with density $\rho = \rho_D \ge \alpha/2$; this description is used only in the analysis. Since $2^{-d_H} \le 2^{-(\ell - r_{\mathrm{s}})} < \alpha/8$ and $d_H s \le \ell r_{\mathrm{s}}$, Lemma 10 gives probability at least $2^{-\ell r_{\mathrm{s}}}(\alpha/4)^2$ of an answer in Z'. On this event, $a_O \in Z'$ and is legal. Because π preserves the first coordinate, $a_O \in \pi\mathcal{A}$. The first player's independent uniform choice from (20) hits a suitable preimage

with probability at least 2^{-r_s}. Averaging over useful data proves (16) with the fixed positive constant

$$\gamma = \gamma_1 2^{-\ell r_s} (\alpha/4)^2 2^{-r_s}. \tag{23}$$

Both strategies use only their own messages. The agreement bound concerns the original unconditioned experiment.

5 Clean coordinates force disagreement

Proposition 11 (Clean coordinates)**.** *If* $\mathrm{OPT}(\mathcal{E}) \le 4/5$*, in particular in a NO instance with* $\xi < 1/100$*,* every *pair of strategies that use only their own messages* (15) *and private randomness, and return legal answers, satisfies*

$$\Pr[\pi a_U = a_O] \le \exp\Big[-\frac{2^{-\dim W - r_s}}{3600} k^{1/3}\Big].$$

Proof. By (2), the equation–variable game has value at most 14/15.

Fix A and combine the visible advice as

$$D_O = (T_O, S_O), \qquad E_A = W \oplus \operatorname{im} A.$$

Conditional on questions and projections, its original intercept and all coefficient columns are independent uniform elements of E_A. Their law does not depend on equation identifiers or selected variable positions. Call a position clean when it is a singleton and its advice coefficient is zero. Conditional on A, positions are independently clean with probability

$$p_A = \beta 2^{-\dim W - \operatorname{rank} A}, \qquad \Pr[e_i = e, j_i = j, i \text{ clean} \mid A] = \omega_e \frac{\beta}{3|E_A|}.$$

Conditioning on cleanliness thus preserves exactly the base game's question-pair law.

Let I be the clean set. Fix only the following background: A, I, the *original intercept on* O, and all questions, projection choices, and coefficients outside I. Do not fix a certificate, output, success event, or answer randomness. For each background of positive probability, the clean question pairs remain independent samples of the base game. This follows from the displayed identity, independence across positions, and independence of the original intercept. Repeated identifiers do not change the product law; positions have distinct local answer coordinates.

Each player can reconstruct its own full message from the background and its own clean questions. Indeed, pulling a singleton coefficient d_i back to an equation gives slopes $(d_i, 0), (0, d_i)$ or (d_i, d_i). The padded intercept is

$$c_U = c_O + \sum_{\substack{i \text{ singleton} \\ j_i = 3}} b_{e_i} d_i.$$

At a clean position $d_i = 0$, so both slopes and the intercept correction vanish. The equation player can reconstruct without knowing which variable was selected; the projected player can reconstruct without knowing the clean equations. Each runs its original strategy and keeps its clean answers. These are legal strategies for the $|I|$-fold base game. Fixing background merely selects a pair of strategies for this bound; it adds no message to the original experiment. Private randomness is a mixture of deterministic strategies and cannot increase the optimum.

Full agreement implies success in every clean position. Input 3 and (2) give

$$\Pr[\pi a_U = a_O \mid \text{background}] \le (1 - 1/3600)^{|I|}.$$

For $I = \varnothing$ use the bound one. We do not assume independence of the strategies' successes. Since $|I|$ is binomial conditional on A,

$$\begin{aligned}\Pr[\pi a_U = a_O] &\le \mathbb{E}_A(1 - p_A/3600)^k \\ &\le \left(1 - \frac{\beta 2^{-\dim W - r_{\mathrm{s}}}}{3600}\right)^k \le \exp\left[-\frac{2^{-\dim W - r_{\mathrm{s}}}}{3600} k^{1/3}\right] \longrightarrow 0. \end{aligned} \tag{24}$$

□

This bounds the *same unconditioned experiment* as (16). Once the auxiliary spaces are fixed, sufficiently large k gives a contradiction. Thus, when $\xi < 1/100$, the intermediate game has the gap

$$\mathrm{YES} : \mathrm{val}(\mathcal{H}) \ge 1 - k\xi - p_*/2, \qquad \mathrm{NO} : \mathrm{val}(\mathcal{H}) \le 99/100. \tag{25}$$

6 Removing weights, amplifying, and writing the output

Rounding weights. If there are N_c weighted constraint occurrences, let $Q = \lceil N_c/\tau \rceil$. Start with $\lfloor Q w_e \rfloor$ copies of occurrence e and distribute the remaining copies to the largest fractional parts. Then

$$\sum_e n_e = Q, \qquad |w_e - n_e/Q| \le 1/Q, \qquad \tfrac{1}{2}\sum_e |w_e - n_e/Q| \le \tau.$$

The weight differences sum to zero. Hence the weight of every subset of constraints changes by at most τ, uniformly over labelings. The optimum changes by at most τ as well. No large common denominator is needed.

A simple bipartite graph. Replace every copy by a path of four edges with three fresh internal vertices. The first three permutations are identities; the fourth is the original constraint. For fixed endpoint labels, all four edges can be satisfied exactly when the original constraint holds. Otherwise exactly three can be satisfied. Distinct paths have disjoint interiors, so their optima can be achieved simultaneously. The resulting game G_{bip} satisfies

$$\left|\mathrm{val}(G_{\mathrm{bip}}) - \left(1 - \frac{1 - \mathrm{val}(\mathcal{H})}{4}\right)\right| \le \tau/4. \tag{26}$$

Place original vertices and path midpoints on one side and the other internal vertices on the other. A loop becomes a four-cycle, and parallel copies have distinct interiors. The graph is simple and bipartite. Reversing an edge leaves a binary translation unchanged.

Parameter order. Choose positive rationals $\varepsilon_0 \le \varepsilon$ and $\delta_0 \le \delta$. If $\tau \le 1/200$, a NO instance has gap at least $1/800$ after subdivision. Choose a fixed integer t such that

$$\left(1 - \frac{1}{10\,240\,000}\right)^t \le \delta_0. \tag{27}$$

This choice precedes the alphabet: the bound in (1) does not depend on it. The denominator is $16 \cdot 800^2$. Next choose rational budgets

$$p_* \le \varepsilon_0/t, \qquad \tau \le \min\{1/200, \varepsilon_0/t\}.$$

The remaining order is

$$p_* \longrightarrow r_* \longrightarrow \eta \longrightarrow \alpha, r_{\mathrm{s}}, \ell_0 \longrightarrow \ell = 3d \ge \ell_0,\ \mathcal{V}, K, W \longrightarrow k, \beta \longrightarrow \xi.$$

Here ℓ_0 is the largest of Input 2's threshold for ℓ at retention $\eta/4$, the threshold for (14), and $3d_0$, where d_0 is such that every $d \geq d_0$ meets the gadget's genericity and $3d \geq r_0$ requirements; it exists because the proportion of nongeneric subspaces tends to zero. Choose a cube $k \geq k_0$, with k_0 from Proposition 5, so large that the bound (24) is below γ; this is possible because $k\beta = k^{1/3} \to \infty$. Then the extraction and clean coordinate bounds contradict each other in the NO case. As in the source, k is a cube so that $\beta = k^{-2/3}$ is rational; this matters only for an effective search for the constants, which the theorem does not need. All thresholds are uniform over the instance and function family. Finally choose a positive rational

$$\xi < \min\{1/100, \varepsilon_0/(2kt)\}.$$

For fixed target errors, these finite constants can be hardcoded into one machine for every input length; their existence therefore suffices here. We make no uniform computability claim in the error targets.

The final game. In the YES case put $\mathrm{err} = k\xi + p_*/2 \leq \varepsilon_0/t$. Output $G = G_{\mathrm{bip}}^{\otimes t}$. A good labeling in every coordinate and a union bound yield

$$\mathrm{val}(G) \geq 1 - t(\mathrm{err} + \tau)/4 \geq 1 - \varepsilon_0/2 \geq 1 - \varepsilon.$$

In the NO case, (1) and (27) give

$$\mathrm{val}(G) \leq \left(1 - \frac{1}{10\,240\,000}\right)^t \leq \delta_0 \leq \delta.$$

The product graph is simple: a pair of product vertices specifies at most one edge in each coordinate. It is bipartite, and independent uniform base edges give uniform product edges. Its constraints are translations over the fixed alphabet $K^t = \mathbb{F}_2^{\ell t}$.

Polynomial output length. All dimensions and repetition counts are fixed before the input. If the processed parity system has $N_{\mathcal{E}}$ occurrences, there are $N_{\mathcal{E}}^k$ tuples. Each has a fixed number of tables and noise outcomes. Each constraint weight is a product of a fixed number of rationals of polynomial bit length. Keys are computed by fixed-dimensional linear algebra; even comparing every pair of keys is polynomial. The rounding multiplicity is linear in the number of tests, and the final power is fixed. The permutation tables have fixed size relative to the input. Thus the entire explicit game can be written in polynomial time, completing the argument for Theorem 1.

Attribution and changes of presentation. The construction and main proof are from [1]. Its matrix-chart argument credits Barak–Kothari–Steurer [5]; the needed calculation is proved here. The table matches passages of this note with those of [1]; passages not listed are inherited in condensed form.

This note	Source	Relation
§3.1 polar map; §3.2 genericity	§3.1, Lemma 3.2; §3.2, Lemma 3.6	Same statements. The polar map is written as a cross product; a square parametrization replaces square roots of monomials.
§3.5, root step (Lemma 4)	§3.5	Replaced, proved here: copies indexed by automorphisms, one quotient, threshold r_0, output dimensions $3d$. Source: copies indexed by injections, two quotients, threshold $3d+2$, all large dimensions.
Erasure (Lemma 8)	Lemma 5.5	Erased matrices get distinct new values outside K, instead of a random recoloring in K with concentration and counting bounds.
Certificates (Lemmas 7, 10; §§4.1–4.3)	Theorem 5.1, Lemmas 5.5, 5.7, 5.8	Constant output, the case $z = 0$ that the proof of Theorem 5.1 already gives, instead of a general affine target $Mz + u$.
§5; §6	§6; §7	Inherited; §6 only shows that the constants exist, while the source also shows that they are computable.

The changes simplify the presentation and some proof steps; none is a claim of a new theorem or an independent verification of the preprint.

A Lean check

OpenAI's repository [1] contains a Lean 4 formalization of the source theorem, with a Comparator challenge that permits only the standard axioms. We report one run of this check, made on 9 October 2026 at the revision pinned in [1], as an unprivileged user on a four-core x86-64 cloud machine. The check script (`ugc_lean_check.py`) and its complete output logs (`ugc-lean-ugt-check.log` and the directory `ugc_out`) are deposited with this note at doi:10.5281/zenodo.23264902.

Repository	`openai/math`, revision `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`
Tools	Lean 4.34.1 (pinned toolchain); Comparator commit `d03acab` (tag `v4.34.0`)
Command	`lake env comparator ComparatorChallenges/UniqueGamesTheorem.json`
Declaration	`OAI.UniqueGamesTheorem.theorem11`
Axioms	`propext`, `Classical.choice`, `Quot.sound`
Verdict	"Lean default kernel accepts the solution", "Your solution is okay!"

Procedure and result. Following the repository's instructions, our script ran `lake update` (manifest and toolchain unchanged), `lake exe cache get` and the command above in the directory `lean`, then a separate `lake build` of the solution and challenge modules and `#print axioms`. The challenge module states the declaration's type: `BinaryGapReduction` ε δ is nonempty for all real $\varepsilon, \delta \in (0, 1/2)$. Comparator exited with code 0, the build succeeded, and the solution's axioms include no `sorryAx`; only the challenge module uses `sorry`, by design, to state the type. The run took 66 minutes.

Caveats. The sandbox was not enforced: `landrun` started, but in our test a write outside its allowed directory succeeded, and the `systemd-run` guard recommended in Comparator's README was unavailable. The check therefore lacked the recommended isolation. We did not audit whether the formal statement matches Theorem 1. Above all, the run concerns OpenAI's formal development of the source theorem. It does not check this note's modified root construction (Lemma 4), its other changes listed under Attribution, or anything else written here.

References

[1] OpenAI. *The Unique Games Theorem.* Preprint, 23 September 2026, 58 pages. Pinned source revision: GitHub source and paper.

[2] Johan Håstad. *Some Optimal Inapproximability Results.* Journal of the ACM 48(4), 798–859, 2001. Input 1 uses Theorem 5.4 and its proof in the author manuscript.

[3] Subhash Khot, Dor Minzer, and Muli Safra. *Pseudorandom Sets in Grassmann Graph Have Near-Perfect Expansion.* Annals of Mathematics 198(1), 1–92, 2023. Input 2 uses Theorem 1.12 in the second 2018 revision.

[4] Irit Dinur and David Steurer. *Analytical Approach to Parallel Repetition.* Proceedings of STOC, 624–633, 2014. Input 3 uses Corollary 1.2 in the 10 June 2014 author version.

[5] Boaz Barak, Pravesh K. Kothari, and David Steurer. *Small-Set Expansion in Shortcode Graph and the 2-to-2 Conjecture.* Proceedings of ITCS, LIPIcs 124, 9:1–9:12, 2019.

[6] Lance Fortnow (prepared with Claude). *The Unique Games Theorem: An Expository Account of the Proof.* 7 October 2026, github.com/fortnow/AI-Papers.

[7] Ariel Elboim. *The Unique Games Theorem: A Short Self-Contained Proof.* Preprint, 2026, doi:10.5281/zenodo.23265572.